

Giving Faces Their Feelings Back

Explicit Emotion Control for Feedforward Single-Image 3D Head Avatars

— Supplementary Materials —

Anonymous ECCV 2026 Submission

Paper ID #*****

1 Implementation Details

1.1 Training Objectives

We train the geometry modulation branch and the appearance modulation branch jointly with separate objectives, using synchronized emotion supervision.

Geometry modulation loss. The geometry branch takes a speech-driven deformation sequence and a target emotion label, and predicts an emotion-aligned deformation sequence using the original parametric face model. We supervise this branch using losses defined in the parameter and surface domains, enforcing consistency with the ground-truth deformation sequence of the same content and target emotion:

$$\mathcal{L}_{\text{geo}} = \lambda_{\text{param}} \|\tilde{\mathbf{p}} - \mathbf{p}^*\|_2^2 + \lambda_{\text{surf}} \|\tilde{\mathbf{V}} - \mathbf{V}^*\|_2^2, \quad (1)$$

where $\tilde{\mathbf{p}}$ and $\tilde{\mathbf{V}}$ denote the predicted deformation parameters and the corresponding mesh vertices, and $(\mathbf{p}^*, \mathbf{V}^*)$ are the ground-truth values under the same articulation and target emotion.

Appearance modulation loss. The appearance branch injects an explicit emotion signal into the reconstruction pipeline to synthesize emotion-aware appearance. Given a reference image and a target emotion label, we supervise the final rendered output against the corresponding target-emotion frame using a reconstruction loss:

$$\mathcal{L}_{\text{app}} = \|\tilde{I} - I^*\|_2^2, \quad (2)$$

where $\tilde{I}$ is the synthesized image and I^* is the ground-truth frame with the same identity, content, and target emotion.

1.2 Dataset Construction and Statistics

This section provides detailed dataset statistics and construction procedures omitted from the main paper for clarity.

Synthesis–Tracking Pipeline. All synthetic data are generated using our synthesis pipeline. Each video is rendered with known camera parameters and subsequently processed by a unified tracking system to extract consistent FLAME [2] parameters (including expression, jaw pose, and head pose). This ensures that both real and synthetic data share a common parametric representation.

All datasets used for quantitative and qualitative evaluation are strictly held out from training. In particular, anchor emotion sequences used for emotion synchronization are never used during evaluation.

Appearance Configuration. The appearance configuration is designed to emphasize identity diversity and appearance coverage, while placing a weaker constraint on temporal continuity.

We sample videos at 5fps. A total of 8,750 reference identities are used. For each identity, we select a reference clip whose viewpoint is closest to the reference image, and render reference videos under seven expression conditions. This results in 61,250 appearance-focused videos in total.

Geometry Configuration. The geometry configuration prioritizes temporal coherence and stable motion patterns for learning geometry and expression modulation.

Videos are sampled at 30fps. We generate 8,750 videos from 1,250 reference identities under the same synthesis settings as the appearance configuration. Compared to the appearance dataset, this configuration trades identity diversity for longer and temporally consistent motion sequences.

Additional Real and Synthetic Data. To support training and evaluation of Zhang et al. [5], we additionally incorporate real facial video datasets (e.g., VFHQ [4]) and synthetic data produced using their original pipeline. All additional data are processed through the same tracking system to ensure consistent FLAME parameterization across sources.

1.3 Architecture and Training Details

Emotion Representation. Emotion labels are encoded as discrete one-hot vectors. Each non-neutral emotion is represented by a 6D one-hot vector, while the neutral emotion is mapped to the all-zero vector. Positional embeddings are applied, and the resulting representation is projected to a 1024-dimensional embedding whose length matches the Transformer token sequence.

The emotion embedding is concatenated with image tokens along the token dimension and provided as conditioning input to the Transformer.

Geometry Modulation Network. The geometry modulation module adopts the same 4-layer Transformer backbone as used in LAM [1]. Its input is a 103-dimensional vector formed by concatenating FLAME expression and jaw pose

parameters. The module outputs emotion-conditioned geometry features that modulate the driving FLAME sequence.

The loss weight for geometry modulation is set to $\lambda = 1$ for all experiments.

Training Settings. For both LAM [1] and Zhang et al. [5], we follow the original training protocols to ensure fair comparison. This includes reference-image preprocessing, FLAME upsampling ratio, rendering resolution, optimizer choice, batch size, and learning-rate schedules.

When fine-tuning from released checkpoints (e.g., LAM [1]), all weights are initialized from the official models. For our implementation of Zhang et al. [5], we replace the template mesh of FlashAvatar [3] with FLAME while keeping the remaining architecture unchanged. Detailed training configurations, including hardware resources, iterations, and total durations, are summarized in Table 1. Additional implementation details and hyper-parameters are kept identical across baseline and emotion-aware variants unless explicitly stated.

2 Additional Evaluation Details

2.1 Held-Out Evaluation Protocol

All identities, sequences, and motions used for quantitative and qualitative evaluation are unseen during training. In particular, no anchor emotion sequence used for emotion synchronization appears in the evaluation set. This prevents information leakage and ensures fair assessment of generalization.

2.2 User Study Details

We conduct a user study to evaluate the quality of portraits by comparing results produced by different 3D head modeling methods against real data. Specifically, we sample 30 videos from the test set, covering diverse identities, viewpoints, and emotions. We invite 141 participants to rate these videos on a scale from 1 (worst) to 5 (best) with respect to three aspects: image quality, emotion recognizability, and emotion vividness. The evaluation criteria for these metrics are as follows:

- **Image quality:** Evaluate the visual fidelity and overall realism of the videos. Participants are instructed to look for visual artifacts, blurring, unnatural distortions, and lighting/texture inconsistencies. A score of 1 indicates severe visual artifacts and a highly artificial appearance, whereas a score of 5 represents photorealistic quality that is visually indistinguishable from real video footage.
- **Emotion recognizability:** Assess how accurately and easily the intended emotional state can be identified from the facial expressions. Participants should consider whether the target emotion is conveyed without ambiguity. A score of 1 means the emotion is completely unrecognizable or confusing, while a score of 5 indicates that the target emotion is instantly and unambiguously clear.

- **Emotion vividness:** Judge the naturalness, intensity, and dynamic richness of the portrayed emotion. This criterion focuses on the presence of realistic micro-expressions, smooth temporal transitions, and human-like expressiveness rather than just static shapes. A score of 1 reflects stiff, robotic, or rigidly artificial expressions, whereas a score of 5 denotes highly nuanced, authentic, and dynamically convincing emotional performances.

2.3 Additional Results

The supplemental material includes extended qualitative comparisons for self-reenactment, cross-reenactment, and emotion transfer on both synthetic and real images, following the same visualization protocol as the main paper.

Reconstruction and reenactment fidelity. Fig. 3 provides a gallery-style visualization of the reconstruction and reenactment results. Notably, for our methods (LAM[†] and Zhang[†]), we additionally present the visual results under different ablation settings. The qualitative comparisons demonstrate that our approach does not compromise rendering quality compared to the baselines.

Emotion transfer Fig. 1 further demonstrates the robustness of our model for emotion transfer on real-world data. Specifically, we evaluate the performance under two distinct settings: an intra-identity setting (the upper 5 rows), where the reference image and driving video share the same identity, and a cross-identity setting (the lower 6 rows), where the identities differ. It is worth noting that the emotion transfer task is inherently identity-agnostic. The core challenge lies in the effective disentanglement and removal of the source emotion, coupled with the accurate reconstruction of the target emotion dictated by the emotion label. Experimental results demonstrate that our proposed method successfully eliminates the source emotion and faithfully synthesizes the target emotion under both settings.

Ablation study. Fig. 2 shows additional results of ablation study. The qualitative comparisons demonstrate that ablating any individual module leads to a degradation in rendering fidelity. Specifically, omitting the geometry branch introduces geometric inconsistencies (e.g., inaccuracies in the degree of eye opening, and the elevation or depression of the mouth corners), while removing the appearance branch causes textural inconsistencies (primarily manifested as the unnatural appearance or disappearance of wrinkles).

2.4 Continuous Emotion Interpolation

To further demonstrate the generalization capability of our model, we provide additional visualization of emotion interpolation in Fig. 4. Although the network is supervised using only seven discrete emotion categories, the results suggest that the model successfully learns a structured and continuous emotion manifold.

Rather than performing a naive linear blend of output parameters (such as FLAME coefficients or motion sequences), which often leads to "expression collapse" or non-physical distortions, our method interpolates within the learned emotion embedding space. This latent representation effectively conditions the geometry deformation and appearance modulation branches in a holistic manner. As illustrated in Fig. 4, by traversing the path between distinct emotion poles (e.g., from *angry* to *neutral*, or *surprised* to *happy*), our model generates a spectrum of intermediate affective states that were never explicitly present in the training set.

Throughout these continuous transitions, several key properties are maintained:

- **Identity Decoupling:** The fine-grained geometric details and textural features that define the individual’s identity remain remarkably stable, regardless of the emotional intensity.
- **Motion Compatibility:** The interpolated emotional expressions naturally coexist with speech-driven lip movements, confirming that our emotion conditioning is well-disentangled from the primary articulation stream.
- **Geometric-Appearance Synergy:** Both the subtle skin wrinkling (appearance) and the macro-level facial muscle contractions (geometry) evolve in perfect synchrony along the interpolation path.

These extended results further validate that our learned emotion space is not merely a set of discrete anchors, but a smooth, navigable domain capable of high-fidelity, fine-grained affective control for 3D head avatars.

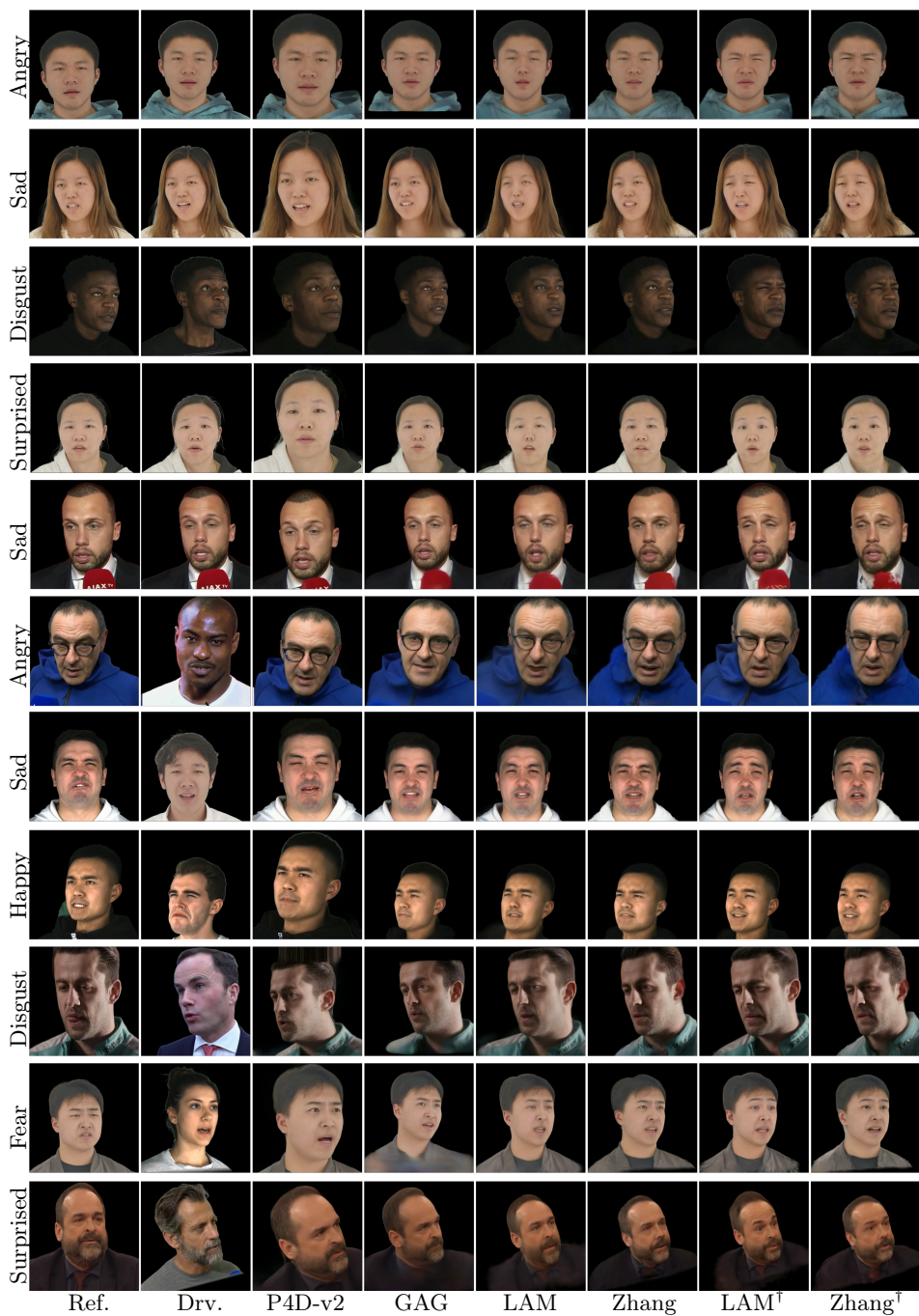


Fig. 1: Emotion transfer on real-world data.

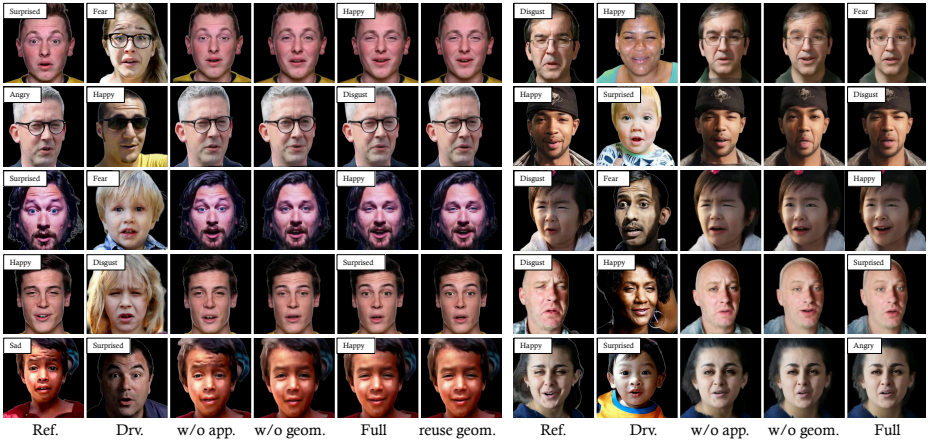


Fig. 2: Additional results of ablation study.



Fig. 3: Reconstruction fidelity galleries on real-world data.



Fig. 4: Emotion interpolation on real-world data.

Table 1: Comparison of training configurations and computational efficiency.

Method	Hardware	Iterations	Training Time
LAM [1]	$8 \times \text{A100 (80G)}^*$	45,500**	~ 2 week*
Zhang et al. [5]	$8 \times \text{A100 (80G)}$	250,000	42.6 h
LAM [†]	$4 \times \text{A100 (80G)}$	30,000	58.4 h
Zhang [†]	$4 \times \text{A100 (80G)}$	50,000	14.8 h

* Training statistics for the original LAM are sourced from official GitHub repository issues.

** Inferred from official LAM-20K checkpoint filenames.

References

1. He, Y., Gu, X., Ye, X., Xu, C., Zhao, Z., Dong, Y., Yuan, W., Dong, Z., Bo, L.: Lam: Large avatar model for one-shot animatable gaussian head. In: SIGGRAPH Conference Papers. pp. 1–13 (2025)

2. Li, T., Bolkart, T., Black, M.J., Li, H., Romero, J.: Learning a model of facial shape and expression from 4d scans. ACM TOG **36**(6), 194–1 (2017)

3. Xiang, J., Gao, X., Guo, Y., Zhang, J.: Flashavatar: High-fidelity head avatar with efficient gaussian embedding. In: CVPR. pp. 1802–1812 (2024)

4. Xie, L., Wang, X., Zhang, H., Dong, C., Shan, Y.: Vfhq: A high-quality dataset and benchmark for video face super-resolution. In: CVPRW (2022)

5. Zhang, J., Chu, L., Li, J., Zang, Z., Li, C., Li, X., Cao, X., Zhu, H., Lu, Y.: Bringing your portrait to 3d presence. In: CVPR (2026)